$$E[E(X|Y,Z)|Y] = \int_{-\infty}^{\infty} E(X|Y,Z)\, f_{Z|Y}(z|y)\, dz$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f_{X|Y,Z}(x|y,z)\, f_{Z|Y}(z|y)\, dx\, dz$$

$$= \int_{-\infty}^{\infty} x\, \frac{f_{XYZ}}{\cancel{f_{Z|Y}} \cdot f_Y}\, \cancel{f_{Z|Y}}\, dx\, dz$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f_{X,Z|Y}\, dx\, dz$$

$$= \int_{-\infty}^{\infty} x\, f_{X|Y}\, dx = E(X|Y)$$

$$\int_{-\infty}^{\infty} f_{X,Z|Y}\, dz = f_{X|Y}$$

PS 3.

1. $M_Z(t) = E(e^{tZ})$ $\qquad f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$

$$= \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(z-t)^2}{2} + \frac{t^2}{2}}\, dz$$

$$= e^{\frac{t^2}{2}}$$

$$M_X(t) = e^{\mu t} e^{\frac{\sigma^2 t^2}{2}}$$